Ansh Tanwar

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SUMMARY

Applied AI engineer who takes LLM systems from prototype to production and owns the outcome. Shipped a real-time voice tutor (133K users, sub-second latency) and a GPT-4o Hinglish calling agent that handles ~50% of customer interactions at Rs 2.8/min, gated by golden-dataset evaluation and A/B tests. Fine-tunes BERT, RoBERTa, ALBERT, and Llama 3 with PEFT / Unsloth (**0.88 F1**; **30%** faster inference). Strong in evaluation, multi-agent orchestration, RAG, and reliability. Amazon ML Nationals winner, Kaggle Master, upstream LiveKit contributor.

WORK EXPERIENCE

Software Development Engineer (AI/ML), Adda247 Mar 2025 – Present

- Own two production voice AI systems end-to-end (architecture, evaluation, deployment, on-call); design adopted across **three Adda Education apps**.
- **Reevo, real-time voice tutor:** Architected on LiveKit, serving **133K unique users** and 190K+ API calls in 3 months; cut round-trip latency from **2000ms to under 1000ms**.
- **LLM memory / context engineering:** Built dual-layer memory (Redis for live session state, vector DB pipeline for long-term user context), **doubling Day-1 retention from 20% to 40%+**.
- **Multi-agent orchestration:** Designed a swarm of specialized agents (Grammar, Tone, CEFR) that lifted weekly engagement from **20 to 40 min/user in 5 weeks**, validated by A/B testing.
- **Pulse AI, autonomous Hinglish calling agent (GPT-4o):** Shipped a 3-agent telephony system handling **~50% of customer call and chat interactions** (thousands of calls/day) at **~60% autonomous resolution** and **Rs 2.8 per minute**, deflecting tier-1 sales calls from human agents.
- **Evaluation:** Gated agent changes against a **hand-labelled golden dataset** (subset used for regression checks) alongside task-level quality metrics, and A/B tested in production.

Automation and AI Engineer, Hike Private Limited Dec 2024 – Mar 2025

- Built an AI Google Meet bot (recording, multilingual transcription, structured notes) adopted company-wide, including the CEO office, engineering, and HR.
- Shipped autonomous workflow agents (FastAPI + Zapier) that cut operational overhead by **70%**.

Data Scientist, Yirifi.ai Pte. Ltd. May 2024 – Sep 2024

- **Fine-tuning:** BERT, RoBERTa, ALBERT, and Llama 3 (PEFT, Unsloth quantized) plus Sentence Transformers on Hugging Face / PyTorch for regulatory classification (**0.88 F1**); cut production NLP inference **30%**.

Undergraduate Researcher, Netaji Subhas University of Technology Dec 2023 – Jun 2024

- **EV charging forecast:** weather + traffic features for residential session energy and duration; RF energy MAE **0.33** / RMSE **0.49**, GB duration MAE **0.28** (transformer weaker). Published ICAIA 2025 (Springer AIS).

OPEN SOURCE & HONORS

- **LiveKit Agents, core contributor (merged upstream):** Shipped the Sarvam.ai plugin, natively unlocking Indian-language STT/TTS in the framework.
- **Amazon ML Challenge 2023, Nationals Winner** (classification over 3M+ rows) | **AWS DeepRacer, World Rank 5** (RL reward design) | **Kaggle Master** (50+ datasets/notebooks, 8,000+ downloads).
- **Technical writing (Medium):** deep-dives on voice AI, agent architectures, and LLM infrastructure | **NannyML:** evaluation experiments and docs for “The Little Book of ML Metrics.”

PROJECTS

Group Study: Adversarial Self-Play & LLM-as-Judge Evaluation Framework | Python, GPT models, LLM-as-Judge | GitHub

- **Evaluation harness:** Ablation studies, cross-domain transfer (sales to debt collection), and regression replay before every prompt patch; 5-dimension LLM-as-judge (conversion, ethics, rapport, pressure, naturalness).
- **Asymmetric self-play:** Attacker LLM vs defender sales agent; only the losing agent is patched each round, so ethical refusal can emerge from the reward signal.

Pulse AI: Production Calling Agent (Hinglish Telephony) | GPT-4o, LiveKit SIP, Sarvam STT, Azure TTS, FastAPI, Qdrant, SPLADE

- **Reliability & RAG:** Zero-buffer FastAPI WebSockets with barge-in; Qdrant + SPLADE hybrid search; full failover across STT, LLM, and TTS.

TECHNICAL SKILLS

Languages: Python, SQL, C++
Fine-tuning: PyTorch, Hugging Face, PEFT, Unsloth (quantized Llama), Sentence Transformers, BERT, RoBERTa, ALBERT, Llama 3
LLM & Agents: OpenAI GPT-4o, multi-agent orchestration, RAG, context engineering, LangChain
Evaluation & Reliability: golden-dataset evaluation, LLM-as-judge, eval harnesses, regression testing, A/B testing, provider failover
Speech & Real-Time: LiveKit (SIP/WebRTC), Whisper, Sarvam.ai, Azure TTS, STT/TTS pipelines, WebSockets
Data & Backend: FastAPI, Qdrant, SPLADE hybrid search, vector DBs, Redis, MongoDB, AWS, vLLM, GPU serving

EDUCATION

B.Tech., Electronics and Communication, Netaji Subhas University of Technology, Delhi |
CGPA 8.68 2021 – 2025